SS-T-316
MARCH 3, 1939

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION
FOR
TILE; PARTITION, GYPSUM

This specification was approved on the above date by the Director of Procurement, for the use of all departments and establishments of the Government, and shall become effective not later than December 1, 1939. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS.

A-1. There are no other Federal specifications applicable to this specification.

B. TYPE.

B-1. Tile shall be of one type.

C. MATERIAL AND WORKMANSHIP.

C-1. Tile shall consist of set gypsum with or without aggregates. The weight of combustible materials shall not exceed 15 percent of the weight of the dry tile.

C-2. Tile shall be free from cracks and other imperfections which would render them unfit for use. Tile shall have straight and square edges and true surfaces.

C-3. Tile shall be dried before shipment.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. *Surfaces* of tile shall be such that they afford a suitable bond with plaster.

E-2. *Shape.*—Tile shall be rectangular except where special shapes are specified. They shall be solid or cellular, the cells being spaced symmetrically.

448250 O - 42

E-3. *Thickness* shall be uniform and within the following limits, in inches:

Nominal:	*Minimum*	*Maximum*
1½ (furring)	1⅜	1⅝
2 (furring)	1⅞	2⅛
2	2	2¼
3	2⅞	3¼
4	3⅞	4¼
5	4⅞	5¼
6	5¾	6¼

E-4. *Height and length.*—Tile shall be of the length and height specified (see par. I-2), with permissible variations of ¼ inch from the nominal height and ⅜ inch from the nominal length.

E-5. *Shell thicknesses* shall be not less than the following in inches:

Thickness of tile	Side and edge shell thicknesses	
	Circular cells	Elliptical or rectangular cells
3	½	11/16
4	¾	13/16
5	⅞	1⅛
6	½	11/16

The sum of the thicknesses of the two side (face) shells plus the thickness of the center vertical web of 6-inch tile shall be not less than 2 inches for tile having circular cells and not less than 2¾ inches for tile having elliptical cells. When the surfaces of the tile are scored, the scoring shall not reduce materially the thickness of the shell.

E-6. *Compressive strength* in pounds per square inch of gross area shall be not less than 75 when dried to constant weight, nor less than 25 when completely saturated.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. *Sampling.*—At least one percent of the number of tile in a shipment but not less than 5 tiles shall be selected representative of the shipment. Twenty-five percent of the selected tiles but not less than 5 individual tiles shall constitute a sample for test. When a shipment consists of more than one car or carrier load, a sample shall be selected from each car or carrier.

F-2. *Inspection and tests.*—

F-2a. *Dimensional determinations.*—Over-all dimensions shall be measured between opposite faces of the tile, the greatest measurement observed being taken as the dimension. Sectional dimensions of cellular tile shall be taken 1 inch in from the ends of the tile, the least measurement observed being taken as the dimension. Shell dimensions shall be measured from the surface of the cell to the normal surface inside the scoring.

F-2b. *Compressive strength.*—

F-2b (1). Not less than five whole tile shall be tested.

F-2b (2). The specimens shall be dried at a temperature of not less than 70° F. (21° C.) nor more than 100° F. (38° C.) in an atmosphere of not more than 50 percent relative humidity. The specimens shall be weighed at 1-day intervals until the loss in weight is less than 0.5 percent per day. If permitted by the inspector, strength tests may be conducted on specimens that are dried at room temperature; however, in cases of dispute, all specimens shall be dried to constant weight as specified above.

F-2b (3). The tile shall be tested in the position in which they are designed to be used. They shall be capped suitably with calcined gypsum mortar before testing. The test shall be conducted after the capping has set, and the test specimen dried to constant weight as specified above.

F-2b (4). The loading head shall cover completely the bearing area of the tile, and the applied load shall be transmitted through a spherical bearing block of proper design. The speed of the moving head of the testing machine shall be not more than 0.05 inch per minute.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G-1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable.

G-2. ~~*Packing.*—There are no special requirements~~.

G-3. *Marking.*—The manufacturer's "brand" name or identifying mark shall be shown upon each tile.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H-1. The following departmental specifications of the issue in effect on date of invitation for bids shall form a part of this specification.

H-1a. *Army.*—United States Army Specification No. 100-2b, Standard Specification for Marking Shipments.

H-1b. *Navy.*—Navy Department General Specifications for Inspection of Material. (Copies of this specification may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D. C.)

H-1c. *Marine Corps.*—Instructions issued by the Quartermaster's Department.

I. NOTES.

I-1. The requirements for dimensions and strength of this specification conform to those of the Standard Specifications for Gypsum Partition Tile or Block (C-52-33) of the American Society for Testing Materials.

I-2. Purchasers should specify the thickness, height, and length of tile, and dimensions of special shapes (if required). Tile 12 inches in height and 30 inches in length are readily available commercially.

I-3. Cellular gypsum tile, 3 or 4 inches in thickness, may be split and used for furring purposes.

Source: https://assist.dla.mil -- Check the source to verify that th

I–4. It is believed that this specification adequately describes the properties necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If for any particular purpose samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–5. This specification covers only the types, classes, grades, sizes, etc., of the commodity as generally purchased by the Federal Government, and is not intended to include all of the types, etc., which are commercially available.

I–6. An index of Federal specifications may be purchased as noted in the paragraph next below, price to be obtained from the Superintendent of Documents.

I–7. Copies of this specification may be obtained upon application, accompanied by money order, or coupon, or cash, to Superintendent of Documents, Government Printing Office, Washington, D. C. Price 5 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise, as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.